

QINHAN TAN

Email: qinhant6@gmail.com

EDUCATION

Princeton University September 2020 - September 2025
Ph.D in Electrical and Computer Engineering - Advised by Prof. Sharad Malik

Princeton University September 2022
M.A. in Electrical and Computer Engineering

Zhejiang University September 2016 - June 2020
B.Eng. in Computer Science and Technology

WORK EXPERIENCE

Apple September 2025 - Present
Formal Verification Engineer

NVIDIA June 2024 - August 2024
GPU Formal Verification Intern

SiFive Inc. June 2022 - August 2022
Hardware Verification Intern

AREA OF INTEREST

- Applied formal methods, including functional verification and security verification
- Formal tools, including model checking algorithms such as IC3
- Hardware security, including side channel attacks and defenses in computer architecture

PUBLICATIONS

1. Tingzhen Dong*, Kenpeng Wang*, Yuheng Yang, Yu-Wei Fan, **Qinhan Tan**, Thomas Bourgeat, Sharad Malik, Mengjia Yan, “The Interplay of Effective Model Checking and Secure Processor Design,” *47th IEEE Symposium on Security and Privacy (IEEE S&P 2026)*.
2. Yuheng Yang*, **Qinhan Tan***, Thomas Bourgeat, Sharad Malik, Mengjia Yan, “Compass: Navigating the Design Space of Taint Schemes for RTL Security Verification,” *31st ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2026)*.
3. **Qinhan Tan**, Akash Gaonkar, Yu-Wei Fan, Aarti Gupta, Sharad Malik, “SecIC3: Customizing IC3 for Hardware Security Verification,” *Design, Automation & Test in Europe Conference (DATE 2026, Best Paper Award Candidate)*.
4. **Qinhan Tan***, Yuheng Yang*, Thomas Bourgeat, Sharad Malik, Mengjia Yan, “RTL Verification for Secure Speculation Using Contract Shadow Logic,” *30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2025)*.
5. Jinyan Xu, Yangye Zhou, Xingzhi Zhang, Yinshuai Li, **Qinhan Tan**, Yinqian Zhang, Yajin Zhou, Rui Chang, Wenbo Shen, “DejaVuzz: Disclosing Transient Execution Bugs with Dynamic Swapable Memory and Differential Information Flow Tracking Assisted Processor Fuzzing,” *30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2025)*.
6. **Qinhan Tan***, Yonathan Fisseha*, Shibo Chen*, Lauren Biernacki, Jean-Baptiste Jeannin, Sharad Malik, Todd Austin, “Security Verification of Low-Trust Architectures,” *30th ACM Conference on Computer and Communications Security (CCS 2023, Distinguished Paper Award)*.

7. **Qinhan Tan**, Aarti Gupta, Sharad Malik, “Usage-Based RTL Subsetting for Hardware Accelerators,” *41st IEEE/ACM International Conference on Computer-Aided Design (ICCAD 2022)*.
8. **Qinhan Tan***, Zhihua Zeng*, Kai Bu, Kui Ren, “PhantomCache: Obfuscating Cache Conflicts with Localized Randomization,” *27th Network and Distributed System Security Symposium (NDSS 2020)*.
9. **Qinhan Tan**, Seetal Potluri, Aydin Aysu, “Efficacy of Satisfiability-Based Attacks in the Presence of Circuit Reverse-Engineering Errors,” *IEEE International Symposium on Circuits and Systems (ISCAS 2020)*.

(* means equal contribution. † means presentation. ‡ means co-presentation.)

RESEARCH EXPERIENCE

Functional Verification for Secure Computation

Collaboration with Univ. of Michigan, co-advised by Prof. Sharad Malik (Princeton) and Prof. Todd Austin (Michigan)

- Develop a compositional verification scheme that can separate the formal checks for individual modules and interconnection.
- Apply this to a secure architecture that consists of multiple crypto modules such as AES and SHA, effectively speeding up the convergence of formal tools.

Model Checking Security Properties

Advised by Prof. Sharad Malik at Princeton University

- Specialize model checking for security verification
- Exploit model symmetry in security property verification to accelerate state-space traversal in the state-of-the-art model checking algorithm, IC3

Security Verification for Speculative Execution Attacks

Collaboration with MIT, co-advised by Prof. Sharad Malik (Princeton) and Prof. Mengjia Yan (MIT)

- Incorporated architecture insights in developing effective verification models and techniques for detecting speculative execution side-channel attacks in RTL implementations (ASPLOS 2025 paper)

Security Verification of Low-Trust Architectures

Collaboration with Univ. of Michigan, co-advised by Prof. Sharad Malik (Princeton) and Prof. Todd Austin (Michigan)

- Developed an end-to-end proof from software programs to RTL hardware implementations for checking security properties (including against side-channel attacks) for a specialized security architecture (Sequestered Encryption, a low-trust architecture) (CCS 2023 paper, Distinguished Paper Award)

RTL Subsetting

Advised by Prof. Sharad Malik at Princeton University

- Designed, implemented and tested an algorithm to extract the subset of an RTL design that is sufficient to provide a specified subset of its complete functionality (ICCAD 2022 paper)

Logic Locking and Satisfiability Attack

Advised by Prof. Aydin Aysu at NC State University

- Evaluated the impact of reverse-engineering errors on the efficacy of Satisfiability Attack that can recover integrated circuit under logic locking (ISCAS 2020 paper)

Countermeasure Against Cache Timing Side Channels

Advised by Prof. Kai Bu at Zhejiang University

- Designed, implemented and evaluated a secure cache design based on dynamic random mapping against cache timing attacks (NDSS 2020 paper)

TEACHING EXPERIENCE

Princeton University	
(ECE 580) Advanced Topics in Computer Engineering: Hardware Security	
<i>Assistant in Instruction</i>	<i>2025 Fall, 2023 Spring</i>
(ECE 206) Contemporary Logic Design	
<i>Assistant in Instruction</i>	<i>2023 Fall, 2022 Fall, 2021 Fall</i>
Zhejiang University	
Computer Organization	2019 Spring
<i>Teaching Assistant</i>	

PATENTS

Chinese Patent for Invention	
1. Kai Bu, Qinhan Tan , Zhihua Zeng, "A defense against cache timing channels based on localized randomization", Number CN201910666998.1. (extending PhantomCache)	

SELECTED HONORS

CSAW 2024 Applied Research Finalist	2024
CCS 2023 Distinguished Paper Award	2023
Princeton School of Engineering and Applied Science Award for Excellence	2023
Princeton ECE Departmental Teaching Assistant Award	2023
Zhejiang Provincial Government Scholarship	2019
Zhejiang University First-class Scholarship for Academic Excellence	2019
Zhejiang University Second-class Scholarship for Academic Excellence	2017-2018

MISCELLANEOUS

Programming Languages	C, C++, Python, Java, Verilog, Chisel, $\text{\LaTeX}$
Tools	Gem5, Verilator, VCS, Jaspergold, VC Formal, IC3